

Information Gathering:

LAB:

- Download the application **xmind** from (<https://xmind.app>)
- Using OSINT, gather information about “Telenor Pakistan”
 - Company Name
 - Key Services/Product
 - Scope of Services
 - Website
 - Fully Qualified Domain Name
 - Hosting Country/Company
 - Age
 - IP Address
 - Geo Location of Web Server (longitude/latitude)
 - Technologies
 - Load Balancer / Web Application Firewall (WAF), if any
- Content Delivery Network (if any)
- Sub-domains
- Domain Registrar
- Shared Hosting (if any)

Now where to gather this information?

If you have **Company name**, go to **Wikipedia** for information like “**Key services/Products, Scope of Services and its Fully Quantified Domain Name**”. Now go to the site **Netcraft > Resources > Research tools > Site Report**. Enter the name of the domain like www.telenor.com.pk.

Background

Site title	Just a moment...	Date first seen	September 2004
Site rank	214178	Primary language	English
Description	Not Present		

Network

Site	http://www.telenor.com.pk	Domain	telenor.com.pk
Netblock Owner	Cloudflare, Inc.	Nameserver	sonny.ns.cloudflare.com
Hosting company	Cloudflare	Domain registrar	pknic.net.pk
Hosting country	US	Nameserver organisation	whois.cloudflare.com
IPv4 address	104.18.13.112 (VirusTotal)	Organisation	Unknown
IPv4 autonomous systems	AS13335	DNS admin	dns@cloudflare.com
IPv6 address	2606:4700:0:0:0:0:6812:d70	Top Level Domain	Pakistan (.com.pk)
IPv6 autonomous systems	AS13335	DNS Security Extensions	Enabled

This site will tell you information like **Hosting Country, Company, IP Address** and some other info.

To find **Geolocation**, go to the site named **IP2Location**, put the IP there and it will tell you the Country, Region, City and Coordinates and some other info.

☐ Permalink	https://www.ip2location.com/104.18.13.112	☐ IP Address	104.18.13.112
☒ IP Address	104.18.13.112	☐ Anonymous Proxy	No
☐ Country	United States of America [US]	☐ Proxy Country	United States of America [US]
☐ Region	California	☐ Proxy Region	California
☐ City	San Francisco	☐ Proxy City	San Francisco
☐ Coordinates of City	37.774930, -122.419420 (37°46'30"N 122°25'10"W)	☐ Proxy ISP	CloudFlare Inc.
☐ ISP	CloudFlare Inc.	☐ Proxy Domain	cloudflare.com
☐ Local Time	01 Aug. 2025 03:17 AM (UTC -07:00)	☐ Proxy Usage Type	(CDN) Content Delivery Network
☐ Domain	cloudflare.com	☐ Proxy Type	DCH
☐ Net Speed	(T1) Data Center/Transit	☐ Proxy ASN	AS13335 CloudFlare Inc.
☐ IDD & Area Code	(1) 415	☐ Threat	-
☐ ZIP Code	94107	☐ Last Seen	-
☐ Weather Station	San Francisco (USCA0987)	☐ Provider	-
☐ Mobile Carrier	-	☐ Fraud Score	0
☐ Mobile Country Code	-		
☐ Mobile Network Code	-		
☐ Elevation	14m		
☐ Usage Type	(CDN) Content Delivery Network		

For information about **Technologies** like which technology are they using, use the **Waplyzer extension**.

To find about **Web Application Firewall** and **Load Balancers**, we have built-in tools in Kali Linux.

Type this in terminal to find info about WAF:

wafw00f www.telenor.com (It will tell you if they are using WAF)

```
(root@kali)~[/home/kali]
# wafw00f www.telenor.com

      ( W00f! )
    /---\
   /  _  \
  /  _  \
 /  _  \
/  _  \
\  _  /
 \  _ /
  \  _/
   \_/_/

404 Hack Not Found
405 Not Allowed
403 Forbidden
502 Bad Gateway
500 Internal Error

~ WAFW00F : v2.3.1 ~
The Web Application Firewall Fingerprinting Toolkit

[*] Checking https://www.telenor.com
[+] The site https://www.telenor.com is behind Azure Front Door (Microsoft) WAF.
[~] Number of requests: 2
```

Type this in terminal to find info about Load Balancer:

lbd Telenor.com (It will tell you if they are using Load Balancer)

```
(root@kali)-[/home/kali]
# lbd telenor.com

lbd - load balancing detector 0.4 - Checks if a given domain uses load-balancing.
                                   Written by Stefan Behte (http://ge.mine.nu)
                                   Proof-of-concept! Might give false positives.

Checking for DNS-Loadbalancing: NOT FOUND
Checking for HTTP-Loadbalancing [Server]:
  Microsoft-Azure-Application-Gateway/v2
  NOT FOUND

Checking for HTTP-Loadbalancing [Date]: 10:20:19, 10:20:20, 10:20:20, 10:20:22, 10:20:22, 10:20:23, 10:20:24,
  10:20:26, 10:20:27, 10:20:29, 10:20:30, 10:20:32, 10:20:34, 10:20:35, 10:20:36, 10:20:36, 10:20:38, 10:20:39,
  10:20:39, 10:20:40, 10:20:41, 10:20:42, 10:20:42, 10:20:43, 10:20:50, 10:20:51, 10:20:51, 10:20:53, 10:20:53,
  10:20:54, 10:20:55, 10:20:56, 10:20:57, 10:20:58, 10:20:58, 10:20:59, 10:21:00, 10:21:01, 10:21:03, 10:21:03,
  10:21:04, 10:21:05, 10:21:06, 10:21:06, 10:21:07, 10:21:07, 10:21:11, 10:21:11, 10:21:12, 10:21:12, NOT FOUND

Checking for HTTP-Loadbalancing [Diff]: NOT FOUND

telenor.com does NOT use Load-balancing.
```

Note: There are other tools for cross-checking like **WMtips**, **Webcheck**, **URLscan.io** (This one is more authentic)

Finding Sub-Domains:

Tools for finding Sub-domains:

- Subdomain finder (Web-based)
- Assetfinder (Linux-based)
 - Install it by typing **apt install assetfinder**
- Subfinder (Linux-based)
 - Install it by typing **apt install subfinder**
- Crt.sh (Web-based)
- Amass

LAB:

- Find out sub-domains of **Telenor.com.pk** using tools mentioned above.
- Validate active sub-domains using **httprobe**, **httpx**(requires golang dependency), & **httpx-toolkit**.
- Compare the output(active-domains) using an appropriate tool to build a final list for sub-domains.

Procedure:

1. Find the sub-domains about the target using tools.

e.g: **subfinder -d Telenor.com.pk**

```
smtp.telenor.com.pk
www.apollo.telenor.com.pk
lba.telenor.com.pk
www.telenor.com.pk
api.telenor.com.pk
www.li.telenor.com.pk
speedtest4.telenor.com.pk
www.digitalssc.telenor.com.pk
ssp-wss.telenor.com.pk
www.beta.corportal.telenor.com.pk
www.smarttrack.telenor.com.pk
kryptonu.telenor.com.pk
www.supplierpartners.telenor.com.pk
www.vpn.telenor.com.pk
0.telenor.com.pk
m.telenor.com.pk
ls.telenor.com.pk
calldropapp.telenor.com.pk
callblocker.telenor.com.pk
www.iba.telenor.com.pk
partner-dpdp.telenor.com.pk
[INF] Found 193 subdomains for telenor.com.pk in 4 seconds 932 milliseconds
```

2. Copy these sub-domains and save them in a file. Suppose filename is sub-domains.txt
3. Now verify the active sub-domains using tool **httprobe** using this command:

cat sub-domains.txt | httprobe

4. This will send probe request to every sub-domain and if he gets response he will list active sub-domains.

Lab: Passive Information Gathering using OSINT

- Use Shodan to extract following information:-
 - All “FortiGate” devices, Country=Pakistan, Organization=Pakistan Telecommunication Company Limited
 - All “Apache” servers, Country=Pakistan, Organization=Pakistan Telecommunication Company Limited
 - All devices, Organization=Telenor Pakistan (Pvt) Ltd
 - NTP Server, Organization=Telenor Pakistan (Pvt) Ltd
 - All devices with OS=Windows XP
 - All devices with OS=Windows XP, Country=Pakistan
 - Nginx Servers, Organization=Mobilink

Suppose you want to know how many or which devices Ptel has on its premises. This is the procedure:

- Go to Shodan.io
- Search like this **cisco country:pk**. This will tell you how many cisco devices are in Pakistan. Then we can also check devices in specific cities.
- Similarly, type **fortigate country:pk**. It will tell you how many fortigate products are in Pakistan with their IP.

The screenshot shows the Shodan search results for the query 'cisco country:pk'. The interface includes a top navigation bar with 'SHODAN', 'Explore', 'Downloads', 'Pricing', and a search bar containing 'cisco country:pk'. On the left, there are two sections: 'TOTAL RESULTS' showing '2,174' and 'TOP CITIES' listing Lahore (710), Karachi (707), Islamabad (160), Rawalpindi (145), and Gujranwala (82). Below these are 'TOP PORTS' listing 22 (921), 80 (482), 161 (267), 23 (137), and 443 (111). The main content area displays three search results. The first result is for IP '221.132.113.162', identified as 'mppt-4th-gw-01 multi.net.pk' and 'Pakistan, Karachi'. It shows an 'SSH-1.99-Cisco-1.25' key type and a key value. The second result is for IP '154.205.142.38', identified as 'JCG JHR-N835R - 首页' and 'Pakistan, Karachi'. It shows an 'HTTP/1.1 200 OK' status and a 'Server: cisco-105'. The third result is for IP '38.54.79.917', identified as 'ownCloud' and 'Pakistan, Karachi'. It shows an 'HTTP/1.1 200 OK' status and a 'Server: cisco-CU011.5'.

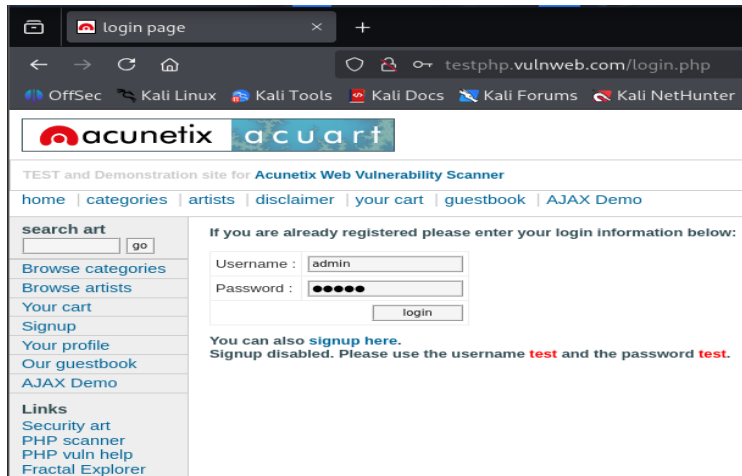
Packet Analysis using Wireshark:

We will be capturing packets from the **Login** page of **Vulnerable web**.

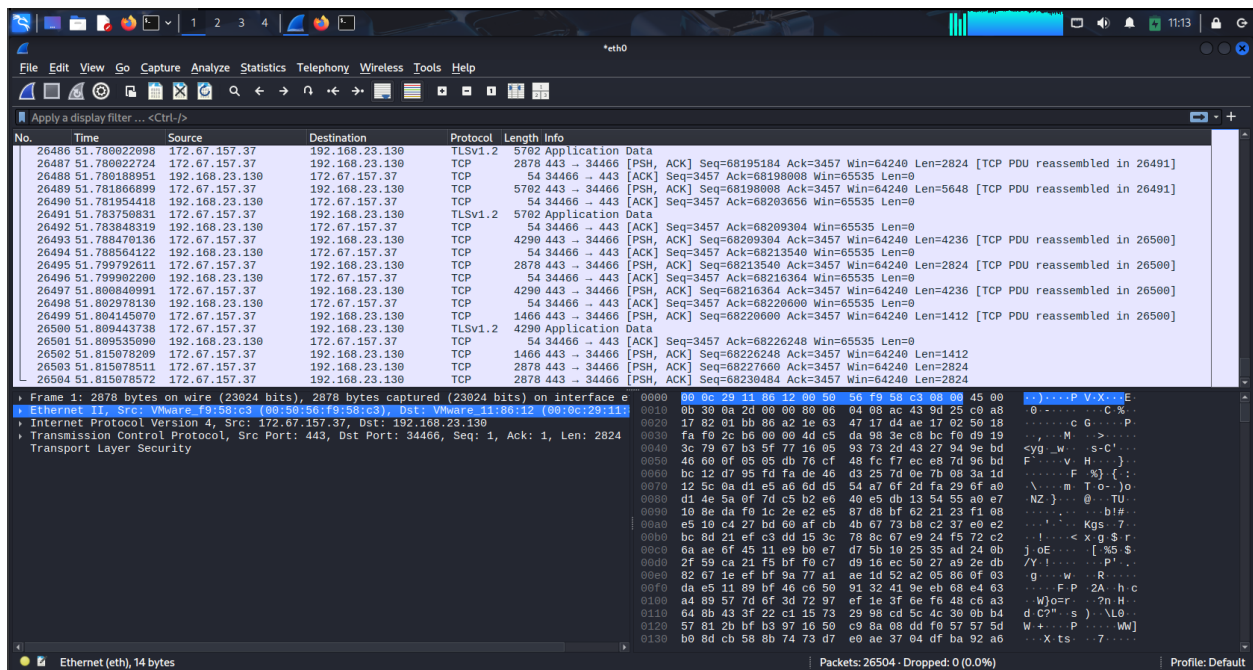
- Open Wireshark and start capturing packets.

The screenshot shows the Wireshark Network Analyzer interface. The top menu bar includes 'File', 'Edit', 'View', 'Go', 'Capture', 'Analyze', 'Statistics', 'Telephony', 'Wireless', 'Tools', and 'Help'. Below the menu is a toolbar with various icons. The main window is divided into three panes. The top pane shows the 'Welcome to Wireshark' message. The middle pane is titled 'Capture' and shows a list of network interfaces. The 'eth1' interface is selected and highlighted in blue. The bottom pane shows a list of capture methods, including 'Cisco remote capture: ciscodump', 'DisplayPort AUX channel monitor capture: dpauemon', 'Random packet generator: randpkt', 'systemd Journal Export: sdjournal', 'SSH remote capture: sshdump', 'UDP Listener remote capture: udpdump', and 'Wi-Fi remote capture: wifidump'.

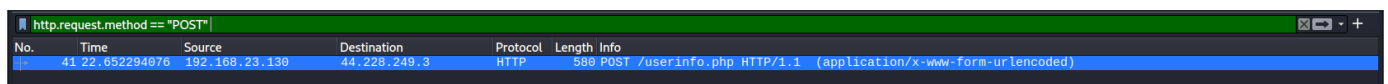
- Go to the login page of the site and enter any credentials.



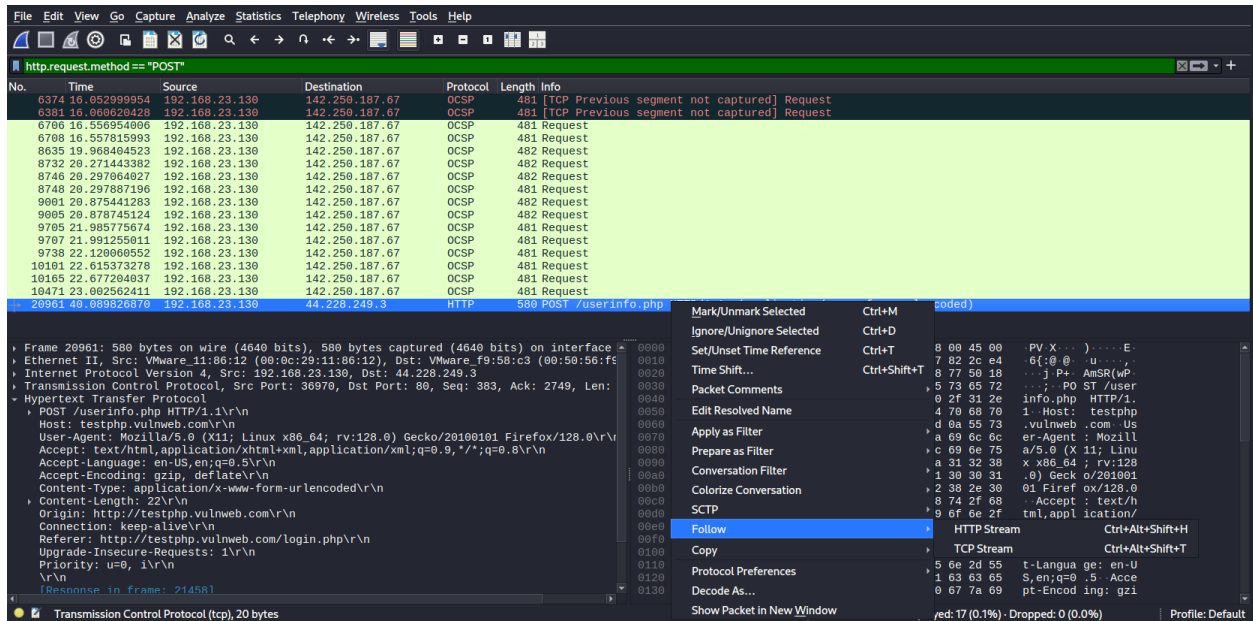
- You will see wireshark will start capturing packets.



- Apply a filter to filter out 'POST' request.



- Right click on it and follow HTTP stream.



- Scroll down until you see credentials.

